CMSC424: Database Design

Module: Introduction/Overview

Instructor: Amol Deshpande
amol@umd.edu

Welcome to CMSC424: Database Design

- Course description
  - *Students are introduced to database systems and motivates the database approach as a mechanism for modeling the real world. An in-depth coverage of the relational model, logical database design, query languages, and other database concepts including query optimization, concurrency control; transaction management, and log based crash recovery. Distributed and Web database architectures are also discussed.*

- Basics
  - Typical in-person instruction, 2 classes a week etc.
  - Class significantly modified from previous iteration
  - Use the discussion forums, office hours as much as possible – assignments are to be done individually, but you are allowed to discuss among yourselves
  - Fill out the survey covering your background, your expectations, and potential issues (on Gradescope)

Logistics

- Instructor: Amol Deshpande
  - 5154 IRB, amol@umd.edu
  - Class Webpage: ELMS
- Email to me: write CMSC424 in the title
  - Only if Piazza not suitable for some reason
  - Piazza allows private posts
- TAs: Hasara Kalumin, Jaechan An

Logistics

- Textbook for reference:
  - Database System Concepts
    - Seventh Edition (6th Edition will be fine too)
    - Abraham Silberschatz, Henry F. Korth, S. Sudarshan
- GitHub:
  - To distribute programming assignments and Slides
  - https://github.com/umddb/cmsc424-spring2026
- Piazza: First resort for any questions
- ELMS: General announcements will be posted there
- GradeScope
  - We will use this for assignments, grading exams, etc.

Grading

- 4 Quizzes – 40%
  - 20-25 mins at the beginning of the class
- Final (25%) – minimum of 50% on this + quizzes combined to pass
- Homeworks and Assignments
  - 4 Homeworks (timed with the quizzes)
  - Programming assignments:
    - Advanced SQL + Programming Languages
    - Web Application Development
    - Implementation Project
  - Late Policy
    - Automatic one-week extension without penalty, but with an explanation of why
    - Beyond that: submissions allowed through the end of the semester with penalties
  - Assignment-0 (Setup) due this Friday

GenAI Tools

- No restrictions on use (except in-class exams)
- In fact, all assignments assume you can use them

CMSC424: Database Design

Module: Introduction/Overview

Background; What We Will Cover

Instructor: Amol Deshpande
amol@umd.edu

The FOUR V's of Big Data

From traffic patterns and music downloads to web history and medical records, data is recorded, stored, and analyzed to enable the technology and services that the world relies on every day. But what exactly is big data, and how can these massive amounts of data be used?

As a leader in the sector, IBM data scientists break big data into four dimensions: **Volume, Velocity, Variety and Veracity**

Depending on the industry and organization, big data encompasses information from multiple internal and external sources such as transactions, social media, enterprise content, sensors and mobile devices. Companies can leverage data to adapt their products and services to better meet customer needs, optimize operations and infrastructure, and find new sources of revenue.

By 2015
4.4 MILLION IT JOBS
will be created globally to support big data, with 1.9 million in the United States

Volume
SCALE OF DATA

40 ZETTABYTES
[43 TRILLION GIGABYTES]
of data will be created by 2020, an increase of 300 times from 2005

2005
2020

It's estimated that
2.5 QUINTILLION BYTES
[2.3 TRILLION GIGABYTES]
of data are created each day

6 BILLION PEOPLE
have cell phones

WORLD POPULATION: 7 BILLION

Most companies in the U.S. have at least
100 TERABYTES
[100,000 GIGABYTES]
of data stored

Variety
DIFFERENT FORMS OF DATA

As of 2011, the global size of data in healthcare was estimated to be
150 EXABYTES
[161 BILLION GIGABYTES]

By 2014, it's anticipated there will be
420 MILLION WEARABLE, WIRELESS HEALTH MONITORS

4 BILLION+ HOURS OF VIDEO
are watched on YouTube each month

30 BILLION PIECES OF CONTENT
are shared on Facebook every month

400 MILLION TWEETS
are sent per day by about 200 million monthly active users

Velocity
ANALYSIS OF STREAMING DATA

The New York Stock Exchange captures
1 TB OF TRADE INFORMATION
during each trading session

Modern cars have close to
100 SENSORS
that monitor items such as fuel level and tire pressure

By 2016, it is projected there will be
18.9 BILLION NETWORK CONNECTIONS
– almost 2.5 connections per person on earth

Veracity
UNCERTAINTY OF DATA

1 IN 3 BUSINESS LEADERS
don't trust the information they use to make decisions

27% OF RESPONDENTS
in one survey were unsure of how much of their data was inaccurate

Poor data quality costs the US economy around
$3.1 TRILLION A YEAR

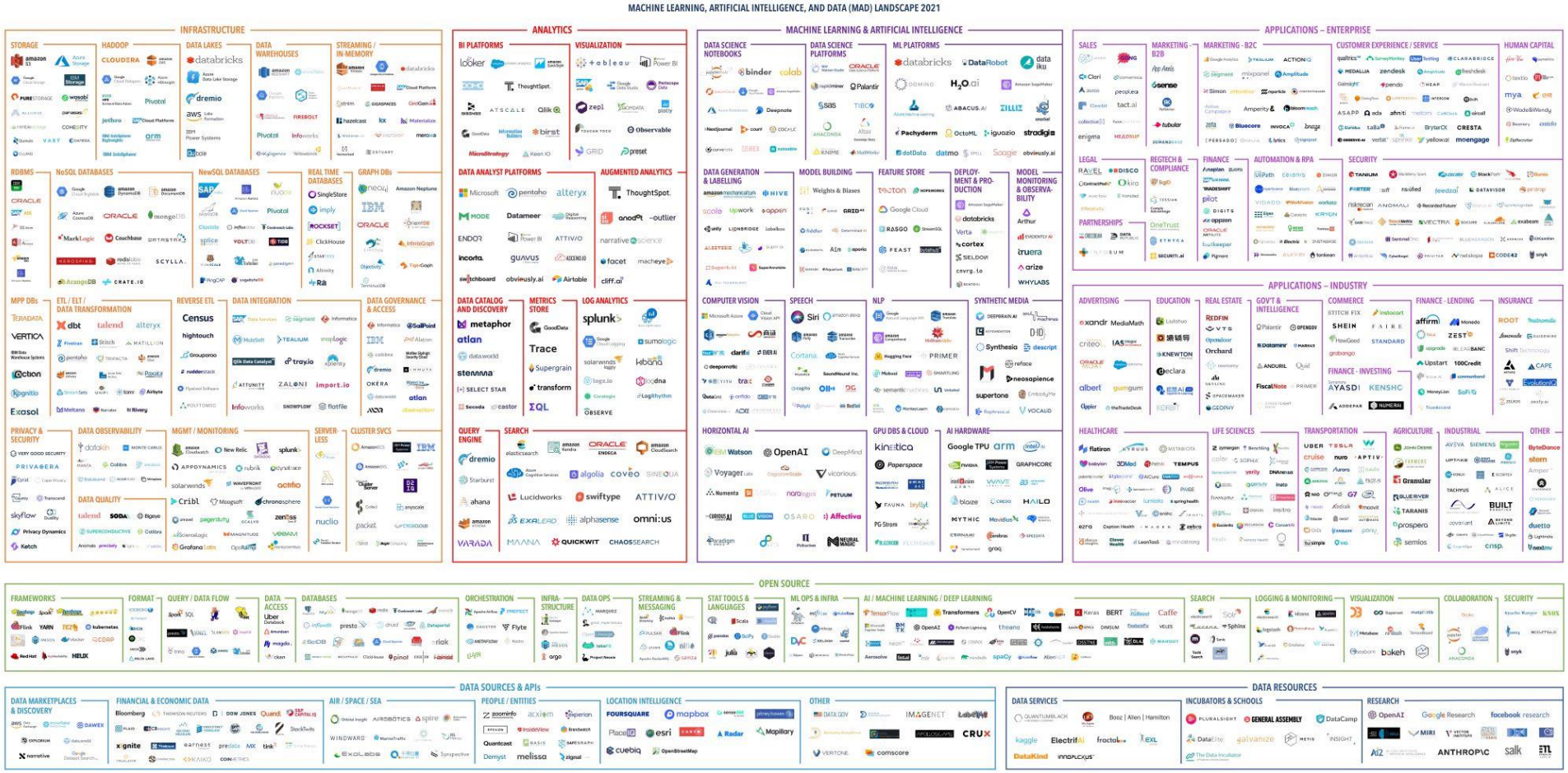
Sources: McKinsey Global Institute, Twitter, Cisco, Gartner, EMC, SAS, IBM, MEPTEC, QAS

IBM

Data Management Systems

Data management systems at the center of most of the new innovative technologies

An incredible amount of ongoing work in building new types of systems



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INFRASTRUCTURE

STORAGE
amazon S3, Azure Storage, Google Cloud Storage, IBM Storage, PURESTORAGE, wasabi, ALLUXIO, panasas, nimblestorage, COHESITY, Qumulo, VAST, DATERA, CLUMIO

HADOOP
CLOUDERA, amazon EMR, Google Cloud Dataproc, Azure HDInsight, HPE Ezmeral Data Fabric, Pivotal, jethro, SAP Cloud Platform, IBM InfoSphere BigInsights, arm, IBM InfoSphere

DATA LAKES
databricks, Azure Data Lake Storage, dremio, aws Lake Formation, IBM Power Systems, Qubole

DATA WAREHOUSES
amazon REDSHIFT, snowflake, Google BigQuery, Azure Synapse Analytics, ORACLE Autonomous Cloud Service, FIREBOLT, Pivotal, Infoworks, Kyligence, Yellowbrick

STREAMING / IN-MEMORY
amazon Kinesis, Google Cloud Dataflow, databricks, confluent, ORACLE, SAP Cloud Platform, striim, GIGASPACES, GridGain, hazelcast, ix, Materialize, Wallaroo.ai, FASTDATA, meroxa, Vectorized, ESTUARY

RDBMS
IBM DB2, ORACLE, SAP ASE, SQLServer, Access, amazon RDS, Amazon Aurora

NoSQL DATABASES
Google Cloud Bigtable, amazon DynamoDB, amazon DocumentDB, Azure CosmosDB, ORACLE, mongoDB, MarkLogic, Couchbase, DATASTAX, AEROSPIKE, redislabs, SCYLLA, ArangoDB, CRATE.IO

NewSQL DATABASES
SAP HANA, Amazon Aurora, nuoDB, MariaDB, Cloud Spanner, Pivotal, Clustrix, influxdata, Cockroach Labs, splice, VOLTDB, TiDB, TIMESCALE, Tratodion, paradigm4, PingCAP, yugabyteDB

REAL TIME DATABASES
SingleStore, imply, ROCKSET, ClickHouse, STARTREE, Altinity, Rill

GRAPH DBs
neo4j, Amazon Neptune, IBM, ORACLE, OrientDB, STARDOG, InfiniteGraph, Objectivity, TigerGraph, TerminusDB

MPP DBs
TERADATA, VERTICA, IBM Data Warehouse Systems, Actian, Kognitio, Exasol

ETL / ELT / DATA TRANSFORMATION
dbt, talend, alteryx, Fivetran, Stitch, MATILLION, pentaho, TRIFACTA, amazon Glue, amazon Athena, Azure Data Factory, Paxata, StreamSets, UNIFI, tamr, Airbyte, Meltano, Narrator, Rivery

REVERSE ETL
Census, hightouch, Grouparoo, rudderstack, Flywheel Software, POLYTOMIC

DATA INTEGRATION
SAP Data Services, segment, Informatica, MuleSoft, TEALIUM, snaplogic, Qlik Data Catalyst, tray.io, xplenty, ATTUNITY, ZALONI, import.io, Infoworks, SNOWPLOW, flatfile

DATA GOVERNANCE & ACCESS
Informatica, SailPoint, IBM, Alation, collibra, McAfee Skyhigh Security Cloud, dremio, IMMUTA, OKERA, WaterLine Data, data.world, atlan, ALTR, ObservePoint

PRIVACY & SECURITY
VERY GOOD SECURITY, PRIVAGERA, Cyral, Cape Privacy, Country, Transcend, skyflow, Duality, Privacy Dynamics, Ketch

DATA OBSERVABILITY
datakin, MONTE CARLO, MANTA, Collibra, datafold, Databand, acceldata, Metaplane

DATA QUALITY
talend, SODA, Bigeye, SUPERCONDUCTIVE, Collibra, Anomalo, precisely, lightup, validio

MGMT / MONITORING
amazon Cloudwatch, New Relic, DATADOG, splunk>, APPDYNAMICS, rubrik, dynatrace, solarwinds, WAVEFRONT by vmware, actifio, Cribl, Moogsoft, chronosphere, unravel, pagerduty, SCALYR, zenoss, ScienceLogic, MAGNITUDE, VEEAM, Grafana Labs, OpsRamp, honeycomb.io

SERVER-LESS
Azure Functions, AWS Lambda, Google Cloud Functions, nuclio, Oracle Fn Project, Oracle Functions Service

CLUSTER SVCS
AmazonECS, IBM Power Systems, IBM, AmazonEKS, Microsoft Cluster Server, D2iQ, Coiled, anyscale, packet, CYCLECLOUD, Bright Computing, Hewlett Packard Enterprise

Data Management

A large fraction of the data still in traditional DBMS systems

Still open and active research areas about improving performance, energy efficiency, new functionalities, changing hardware spectrum (SSDs) and so on…

Much of the data not stored in traditional database systems today

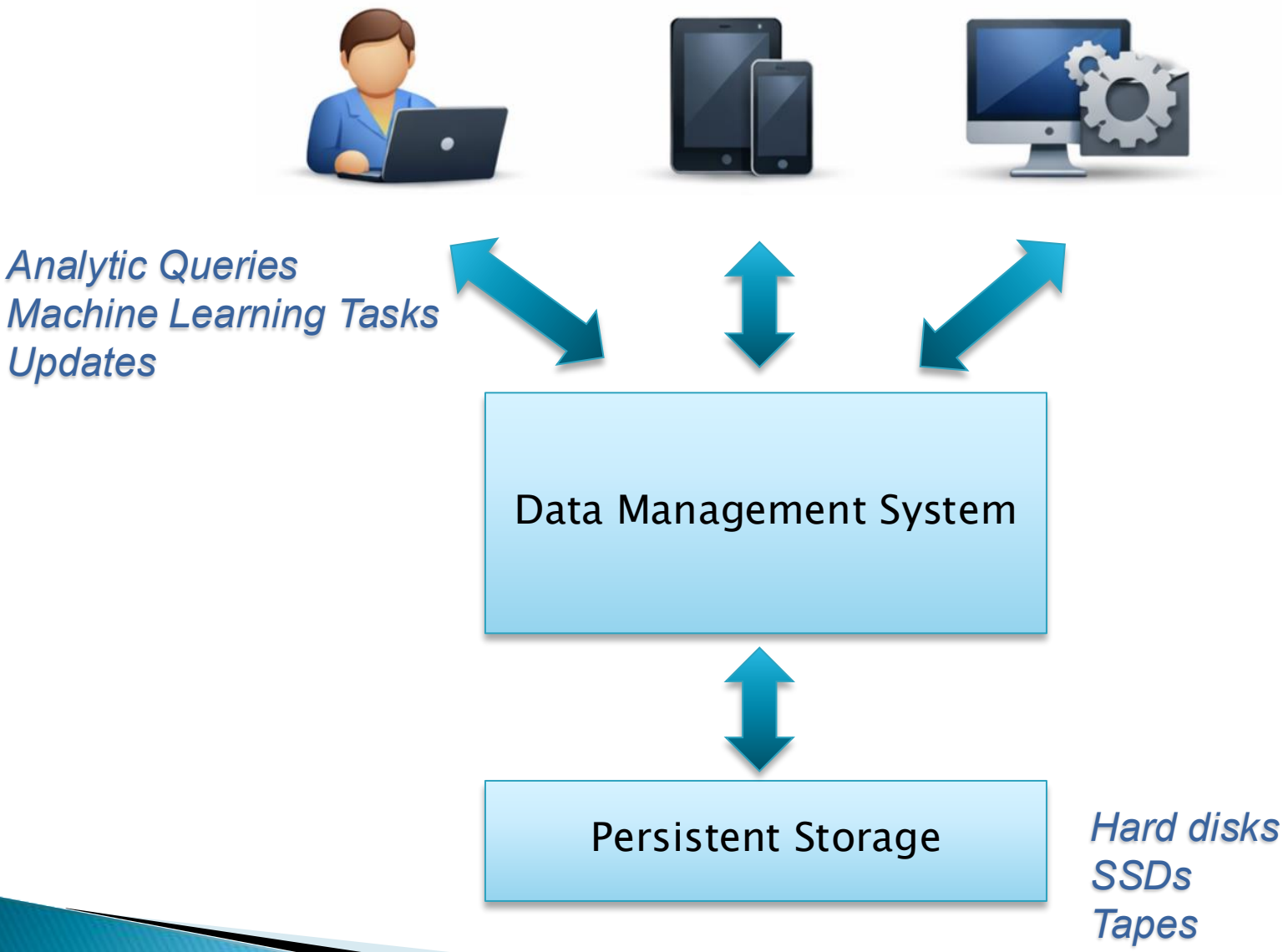
For a variety of fairly valid reasons

- Stream processing systems (focusing on *streaming* data)
- Special-purpose data warehousing systems (most start from some RDBMS)
- Batch analysis frameworks (like Hadoop, Pregel, Spark, …)
  - Typically data stored in distributed file systems
- Key-value stores (like HBase, Cassandra, Redis, …)
  - Basically persistent distributed hash tables
- Semi-structured/Document data stores (for XML/JSON query processing)
- Graph databases
- Data lakes (e.g., scientific data, machine learning data)

However, many lessons to be learned from database research

We see much reinvention of the wheel and similar mistakes being made as early on

What's a Database System?



A Database System

Data Definition Language (DDL)
for specifying the "schema"
Data Manipulation Language (DML)
for specifying queries, updates

- Uses a specific data model
- Hides low-level details
- Provides guarantees about consistency of data, etc.

Data Management System

Persistent Storage

Hard disks
SSDs
Tapes

A Typical "Relational" Database

DDL

```
create table department
    (dept_name    char (20),
     building     char (15),
     budget       numeric (12,2));
```

DML

```
select instructor.name
from instructor
where instructor.dept_name = 'History';
```

Data Management System

Persistent Storage

Hard disks
SSDs
Tapes

Model: Relational

Abstraction of a table

ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

MongoDB (A Document Database)

DDL (Not Enforced)

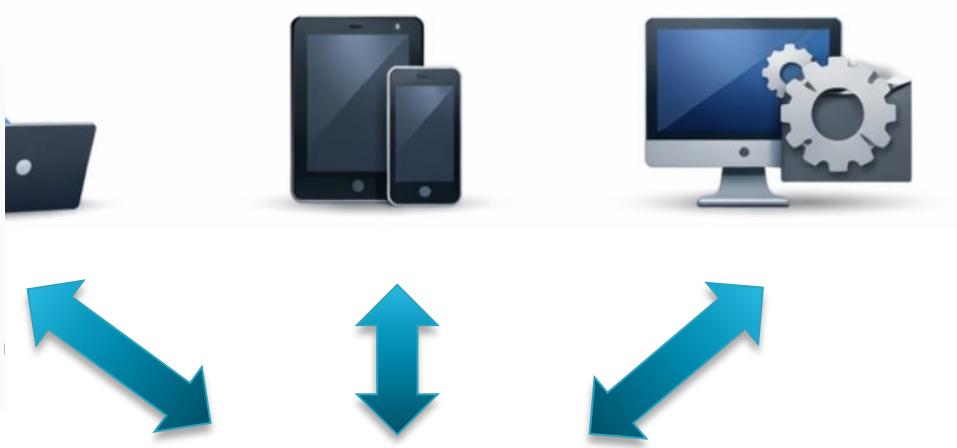
```
$jsonSchema: {
   bsonType: "object",
   title: "Student Object Validation",
   required: [ "address", "major", "name", "year" ],
   properties: {
      name: {
         bsonType: "string",
         description: "'name' must be a string and is
      },
```

DML

```
db.users.find({ status: "active" })
```

Model: JSON

```
{"instructors": [
      {"ID": "22222",
       "name": "Einstein",
       "dept_name": "Physics",
         ....
      },
```



Data Management System

Persistent Storage

Hard disks
SSDs
Tapes

Spark (RDD Interface)

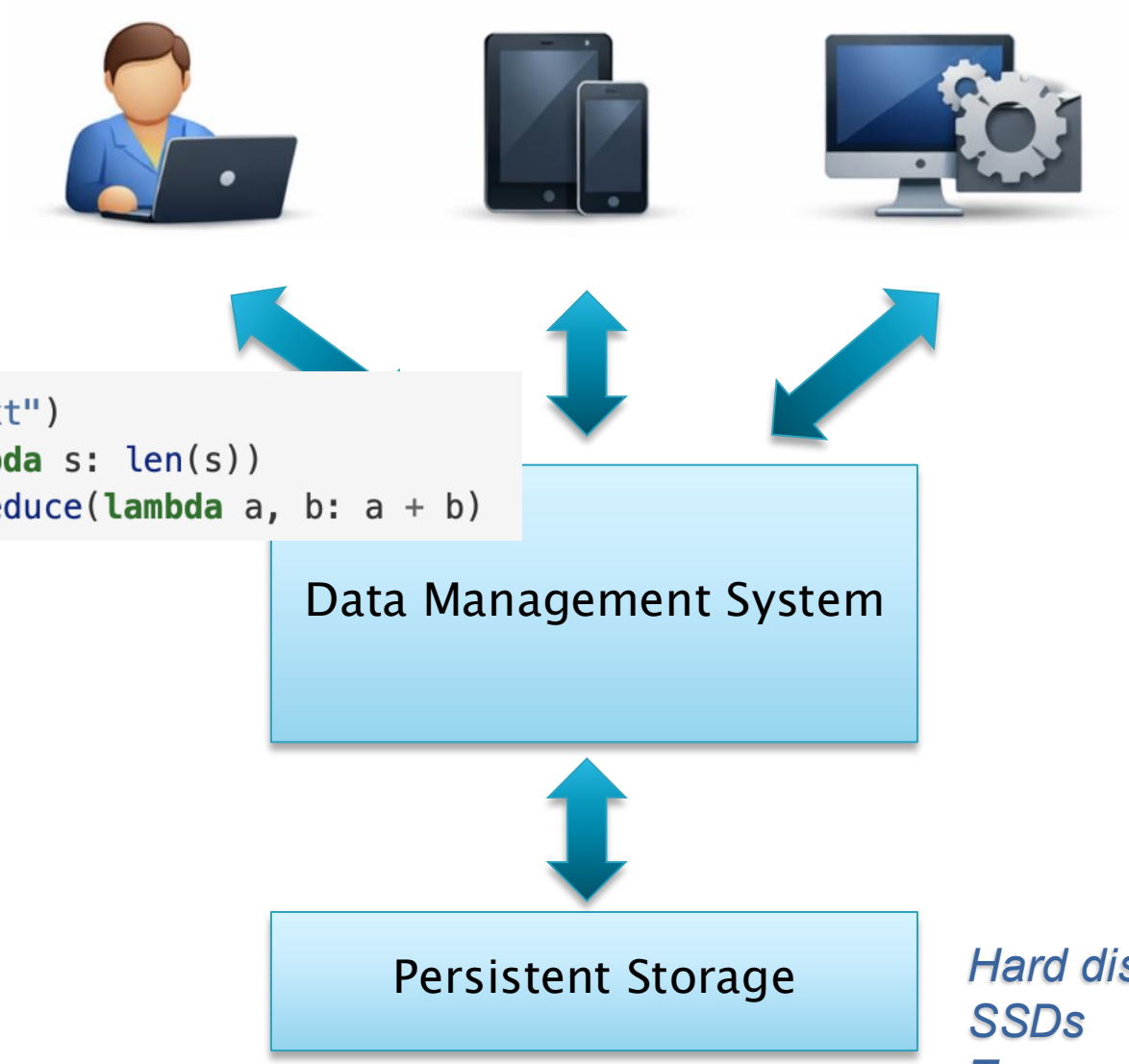
DDL

Schema implicit in the operations

DML

```
lines = sc.textFile("data.txt")
lineLengths = lines.map(lambda s: len(s))
totalLength = lineLengths.reduce(lambda a, b: a + b)
```

Model: Collections of Objects, or Key-Value Pairs



Data Management Challenges (1)

- How to “represent”/”model” the data so users can easily understand, query, analyze, and update?
  - Focus on “logical” representation, not “physical”
  - Should be easy for humans to work with

As tables (like Excel)?

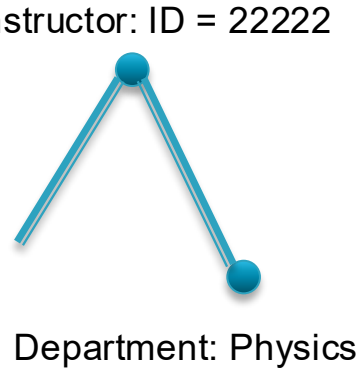
ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000

dept_name	*building*	*budget*
Comp. Sci.	Taylor	100000
Biology	Watson	90000
Elec. Eng.	Taylor	85000

JSON Documents?

```
{“instructors”: [
        {“ID”: “22222”,
         “name”: “Einstein”,
         “dept_name”: “Physics”,
           ….
         },
         ….
```

Graph?



Data Management Challenges (2)

- How to query the data? How to process/analyze it ? (from users'/analysts' perspective)
  - "find the stores with the maximum increase in sales in last month"
    - We can't expect the users to write Java programs
  - "how much time from here to Pittsburgh if I start at 2pm ?"
    - Data is there (GPS, live traffic data), but requires predictive capabilities
  - Needs to be easy to use, but powerful enough for what needs to be done
    - Otherwise users will simply take out all the data and write their own programs

Query Languages and Programming Abstractions

Data Management Challenges (3)

- How to query the data? How to process/analyze it ? (from implementation perspective)
  - How to handle the increasing scale of the data
  - How to support more and more complex analytics/machine learning over the data
  - How to process the data in real-time, as it is arriving into the system
  - How to find the best way to execute a query with least resources/time
  - How to utilize large numbers of machines to execute queries more efficiently
  - …

Database Query Engines
Big Data Systems

Data Management Challenges (4)

- Modifying/updating the data given:
  - Many users simultaneously updating the data…
  - While the data is (potentially) replicated across the globe…
  - And any server or the network may go down at any point.

Update 1	Update 2
a = a + 10	inc = a * 0.15
b = b - 10	a = a + inc
save a & b	b = b - inc
	save a & b

If allowed to run concurrently →

a = a + 10
inc = a * 0.15
a = a + inc
b = b - 10
save a & b
b = b - inc
save a & b

Doesn't happen if forced to run one after the other, but unacceptable performance

Data Management Challenges (4)

- Modifying/updating the data given:
  - Many users simultaneously updating the data…
  - While the data is (potentially) replicated across the globe…
  - And any server or the network may go down at any point.

Update

a = a + 10
b = b - 10
save a
save b

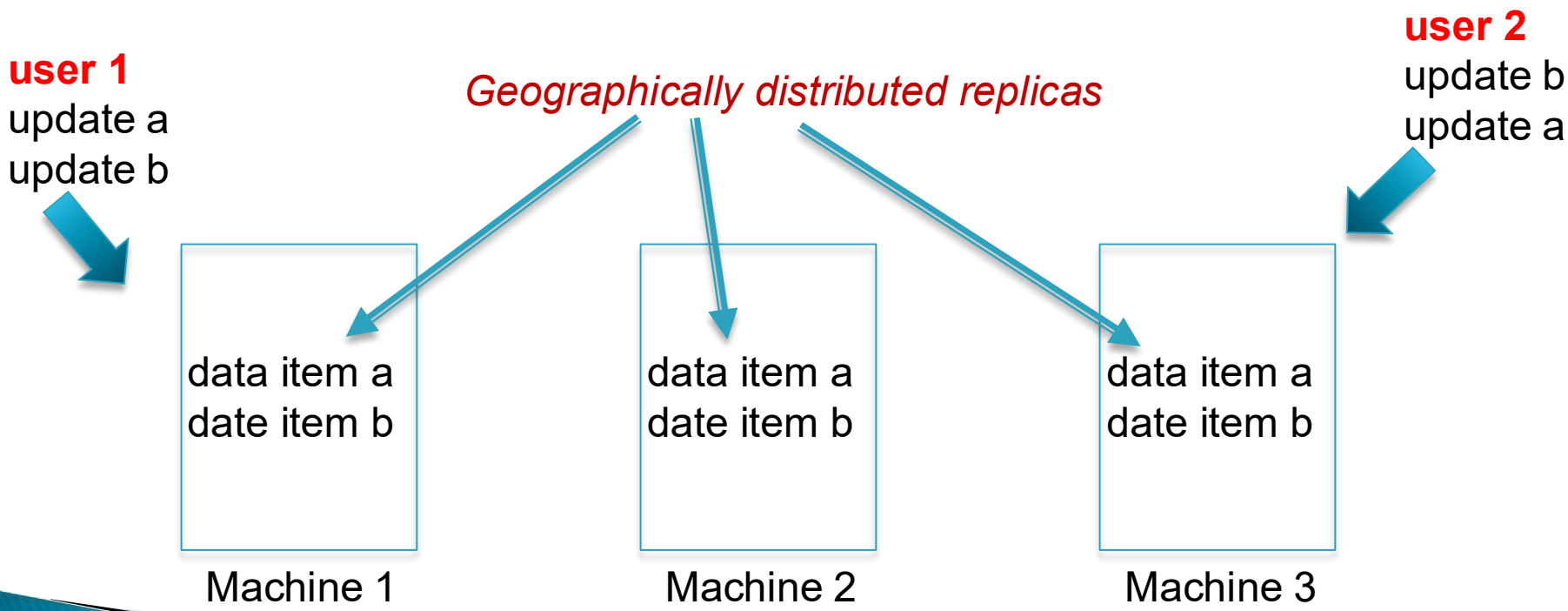
System crashes

Need to figure out how to handle partially finished transactions

More complications if we do update-in-place (i.e., a = a + 10 updates the original value)

Data Management Challenges (4)

- Modifying/updating the data given:
  - Many users simultaneously updating the data…
  - While the data is (potentially) replicated across the globe…
  - And any server or the network may go down at any point.



Data Management Challenges (4)

- Modifying/updating the data given:
  - Many users simultaneously updating the data…
  - While the data is (potentially) replicated across the globe…
  - And any server or the network may go down at any point.
- From Users' perspective:
  - What kinds of guarantees can be provided to the users about the end result, and about the order in which updates are made?
  - What if the users are okay with weaker guarantees?
- From Implementation perspective:
  - How to provide the guarantees efficiently?

Transactions

Data Management Challenges (5-∞)

- Data Integration, Transformation, and Cleaning
  - Large amounts of time spent in just getting data ready for analysis, by some accounts 90-95%
  - Real-world data has a lot of noise, missing values, etc…
- Security
  - Access control, use of encryption, …
- Privacy
  - How to allow users to query data without revealing it
  - How to enable public data releases (e.g., by Census Buereau)
- …

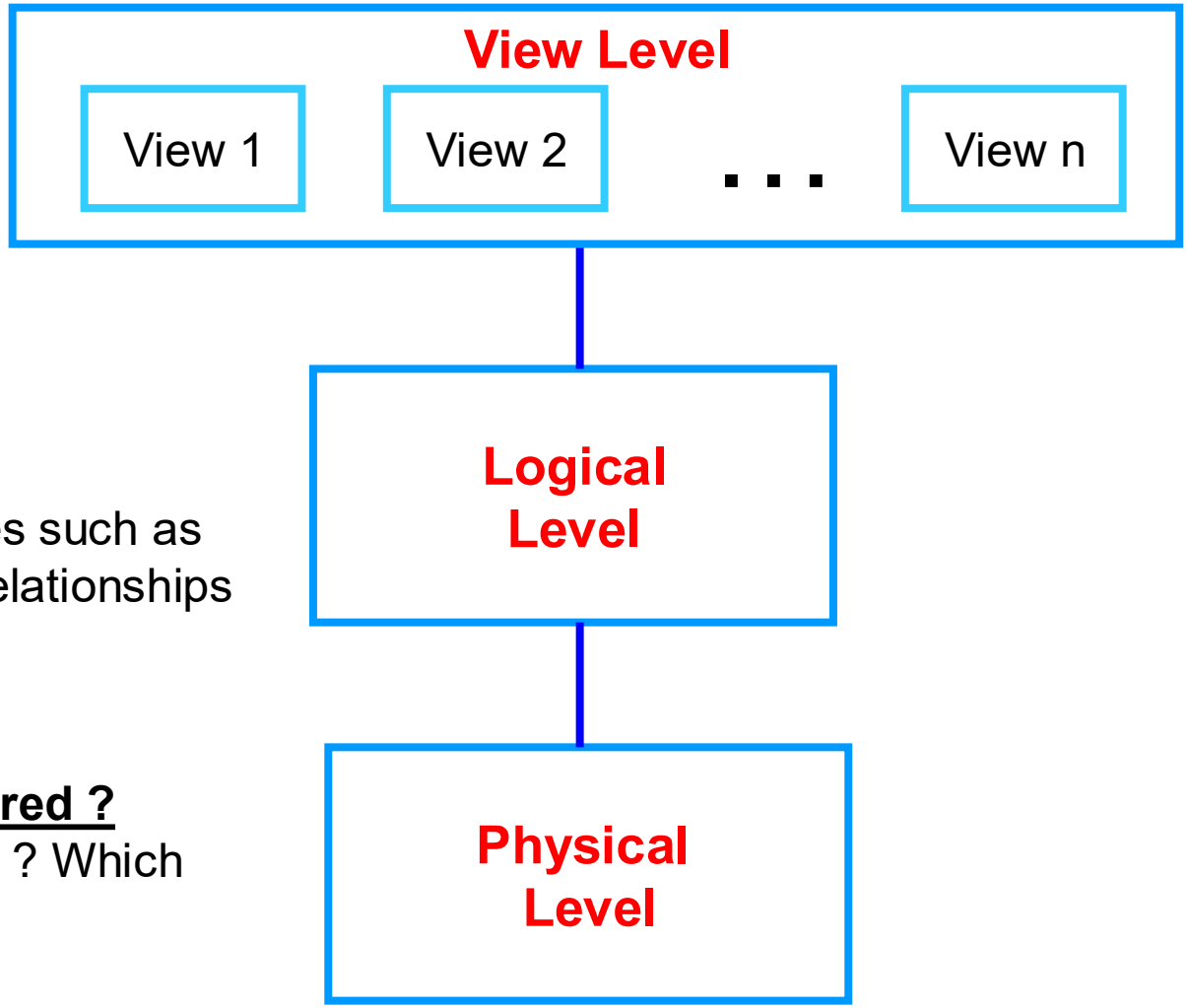
Data Abstraction

- Probably *the* most important purpose of a DBMS
- Goal: Hiding *low-level details* from the users of the system
  - Alternatively: the principle that
    - *applications and users should be insulated from how data is structured and stored*
  - Also called *data independence*

- Through use of *logical abstractions*

Data Abstraction

What data users and application programs see ?



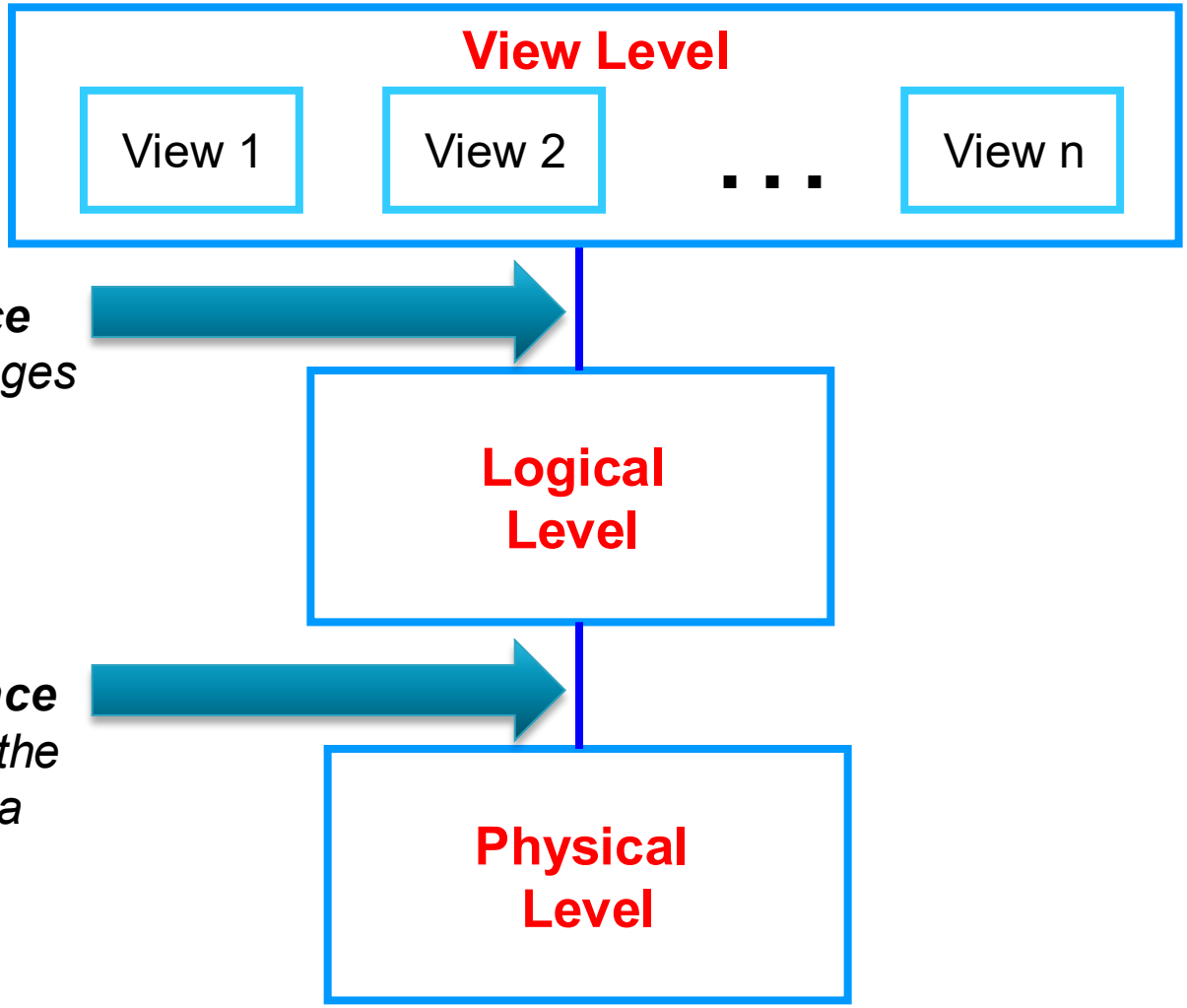
What data is stored ?

describe data properties such as data semantics, data relationships

How data is actually stored ?

e.g. are we using disks ? Which file system ?

Data Abstraction



Logical Data Independence
Protection from logical changes to the schema

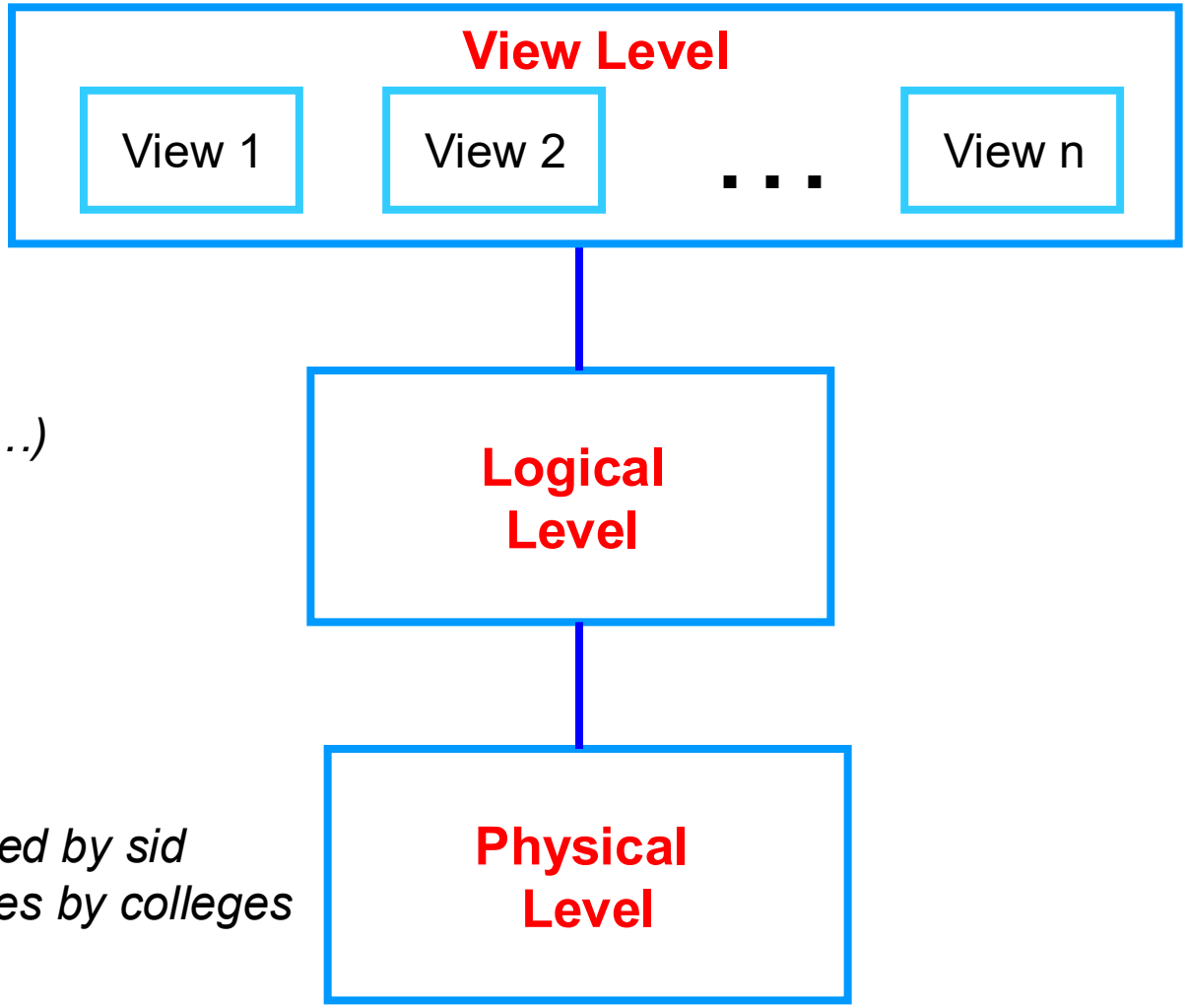
Physical Data Independence
Protection from changes to the physical structure of the data

Data Abstractions: Example

A View Schema
course_info(#registered,…)

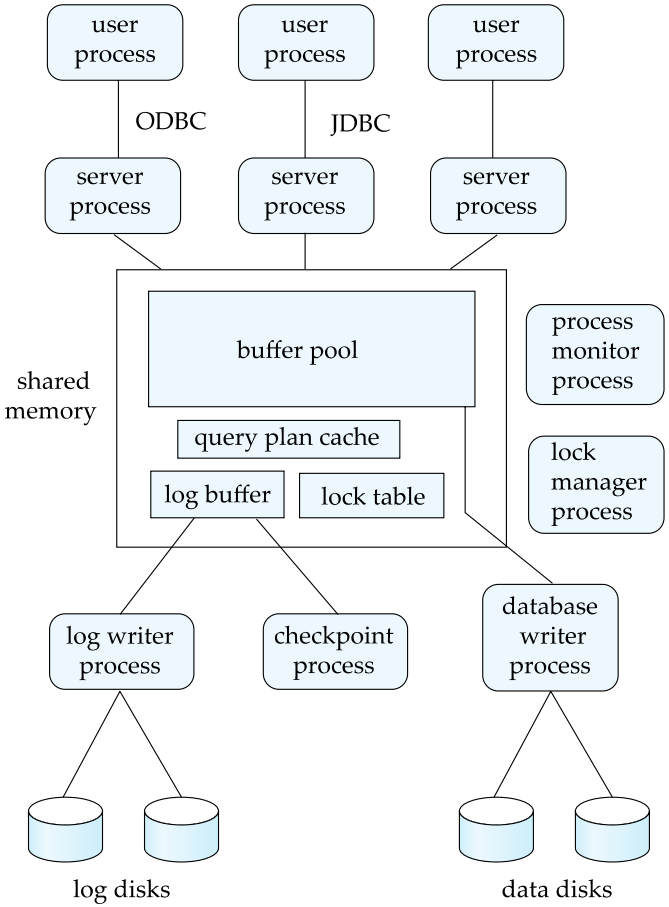
Logical Schema
students(sid, name, major, …)
courses(cid, name, …)
enrolled(sid, cid, …)

Physical Schema
all students in one file ordered by sid
courses split into multiple files by colleges



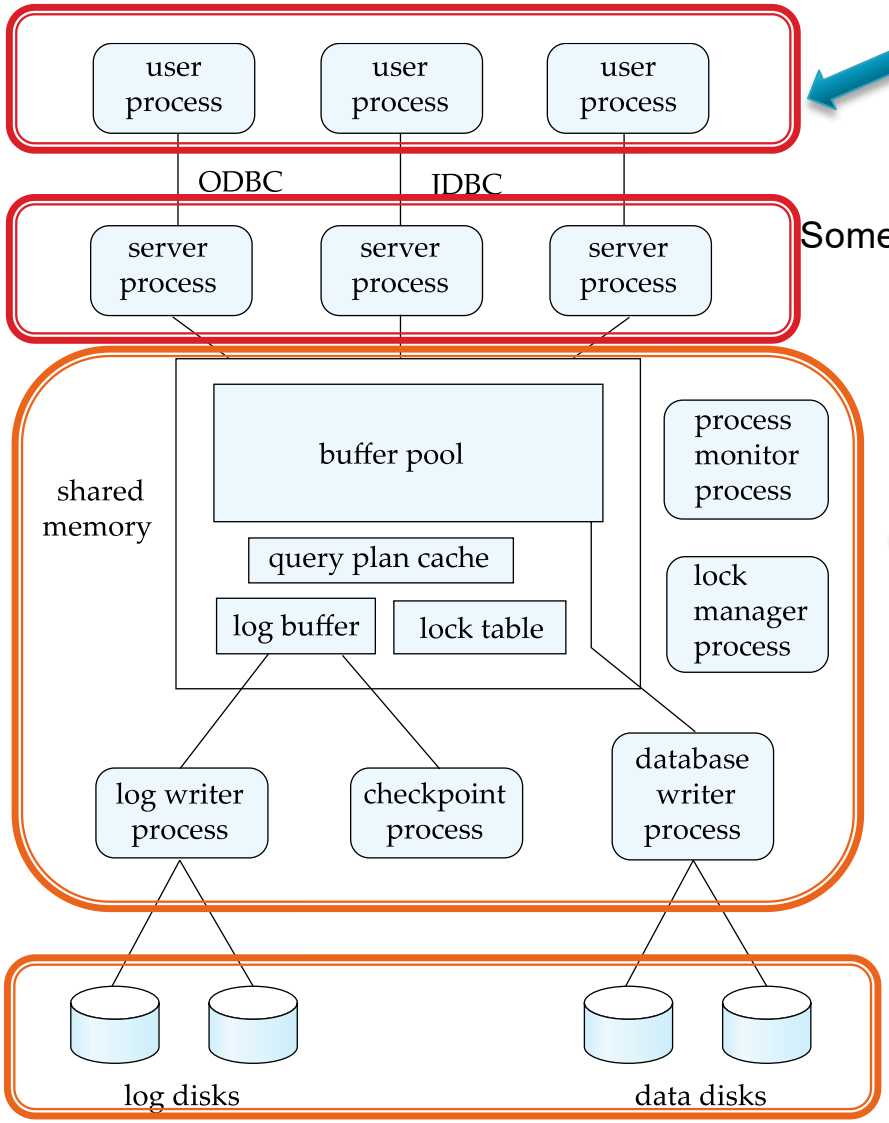
Database Architecture: Pre-2000's

- All data was typically in hard disks or arrays of hard disks
- RAM (Memory) was never enough
  - So always had to worry about what was in memory vs not
- Almost no real "distributed" execution
  - Different from "parallel", i.e., on co-located clusters of computers
- Relatively well-understood use cases
  - Report generation
  - Interactive data analysis and exploration
  - Supporting transactions



From Chapter 20

Traditional RDBMS Architecture



Clients may be anywhere – e.g., ATMs, desktops, laptops, web apps etc.

Talk to the database using standard protocols like JDBC/ODBC, SOAP, or REST (today), or proprietary protocols

Some sort of load balancer or intake mechanism

Typical components in a database system: some for queries, some for transactions

Maybe on a single physical computer or a cluster connected by a fast network

Data Storage Systems:
(1) Punch cards (long time ago)
(2) Hard disks (still prevalent)
(3) SSDs

Need "redundancy" and "fault-tolerance"
Data once stored should always be there

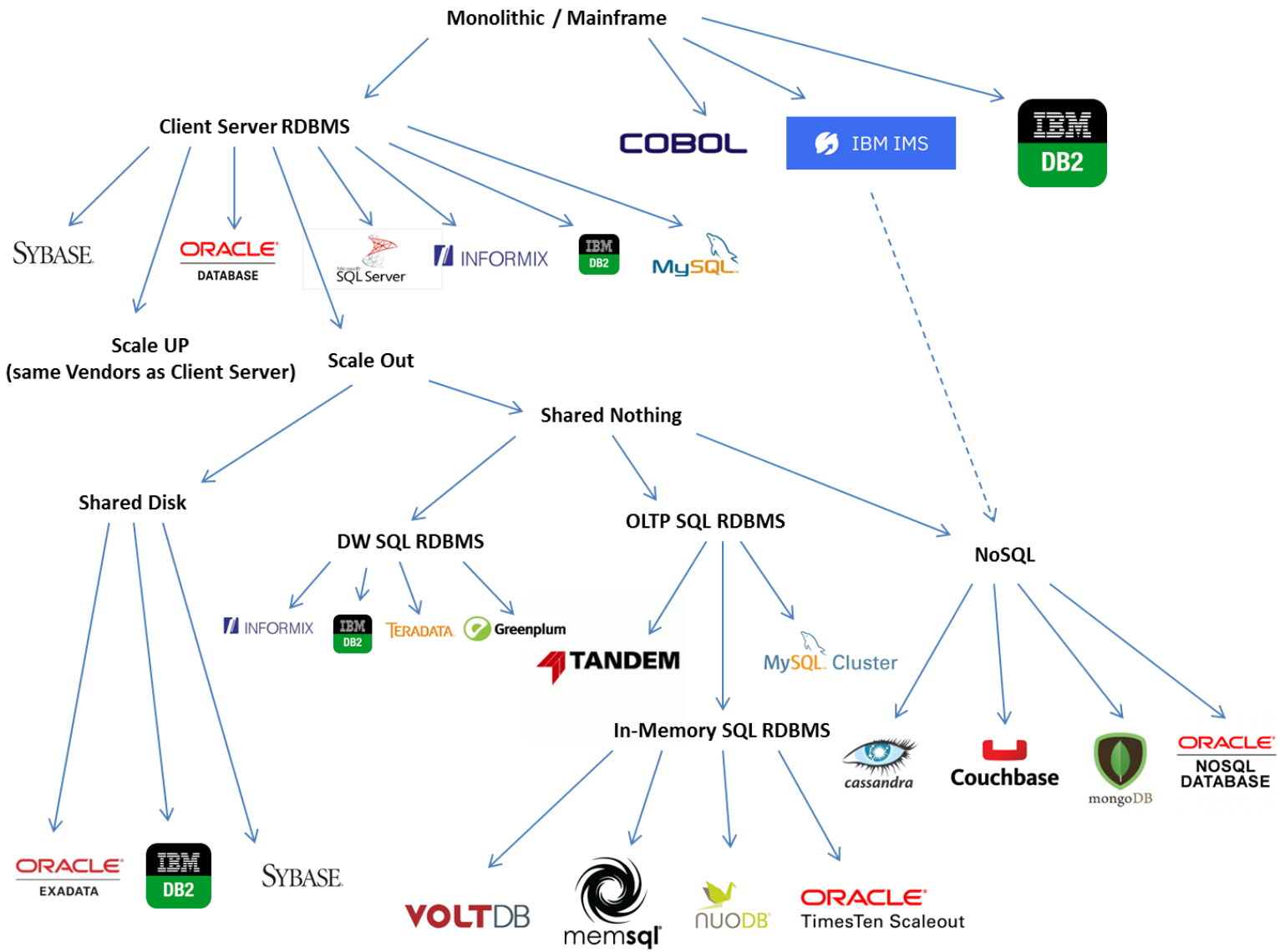
RAID = Redundant Array of Independent Disks

Database Architecture : Today

- Much more diversity in the architectures that we see
  - More modern hardware architectures
    - Massively parallel computers
    - SSDs
    - Massive amounts of RAM – often don't need to worry about data fitting in memory
    - Much faster networks, even over a wide area
    - Virtualization and Containerization
    - Cloud Computing
  - As a result: Data and execution typically distributed all over the place

- Much more diversity in data processing applications
  - Much more non-relational data (images, text, video)
  - Data Analytics/Machine learning more common use-cases

- Much more diversity in "data models"
  - Document data models (JSON, XML), Key-value data model, Graph data model, RDF



From: https://blogs.oracle.com/timesten/the-evolution-of-db-architectures
(Oracle-focused)

Data Warehouses

For: Large-scale data processing (TBs to PBs)

Parallel architectures (lots of co-located computers)

SQL and Reporting

No transactions

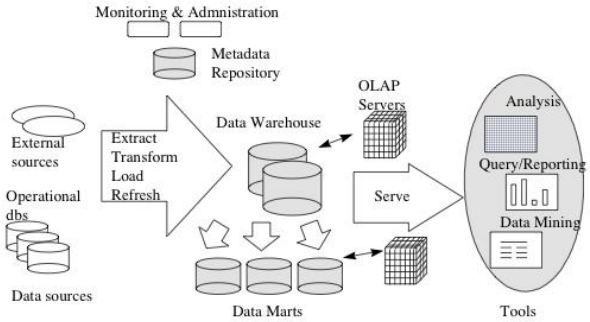


Figure 1. Data Warehousing Architecture

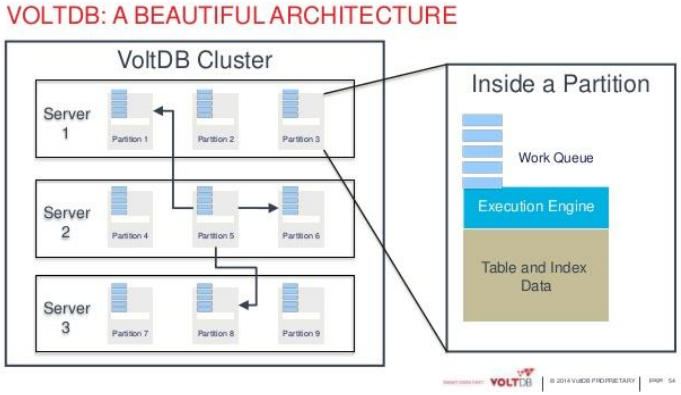
In-memory OLTP (on-line transaction processing)

For: Extremely fast transactions

Many-core or parallel architectures

Very limited SQL – mostly focused on “writes”

Typically assume data fits in memory across servers



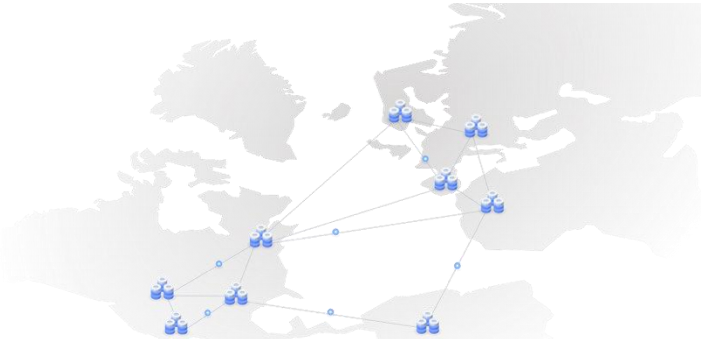
Highly available, distributed OLTP

For: Distributed scenarios where clients are all over the world

Focus on “consistency” – how to make sure all users see the same data

Limited SQL – mostly focused on “writes”

Considerations of memory vs disk less important

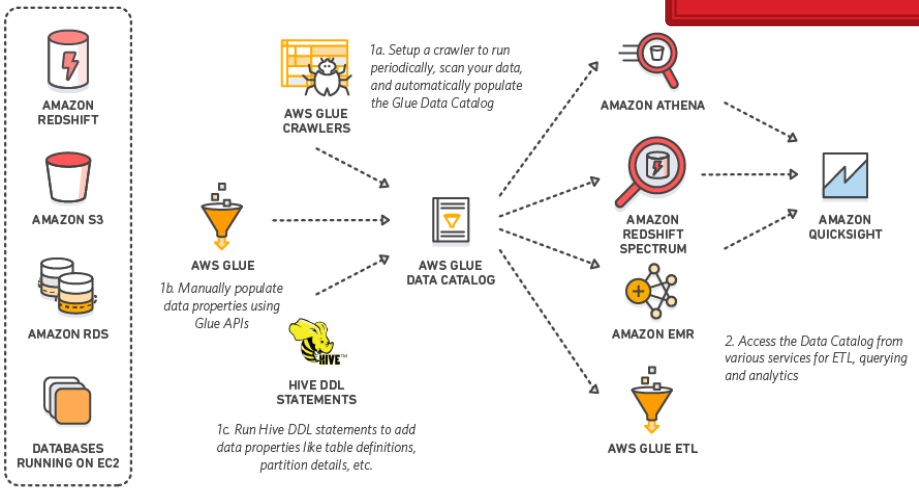


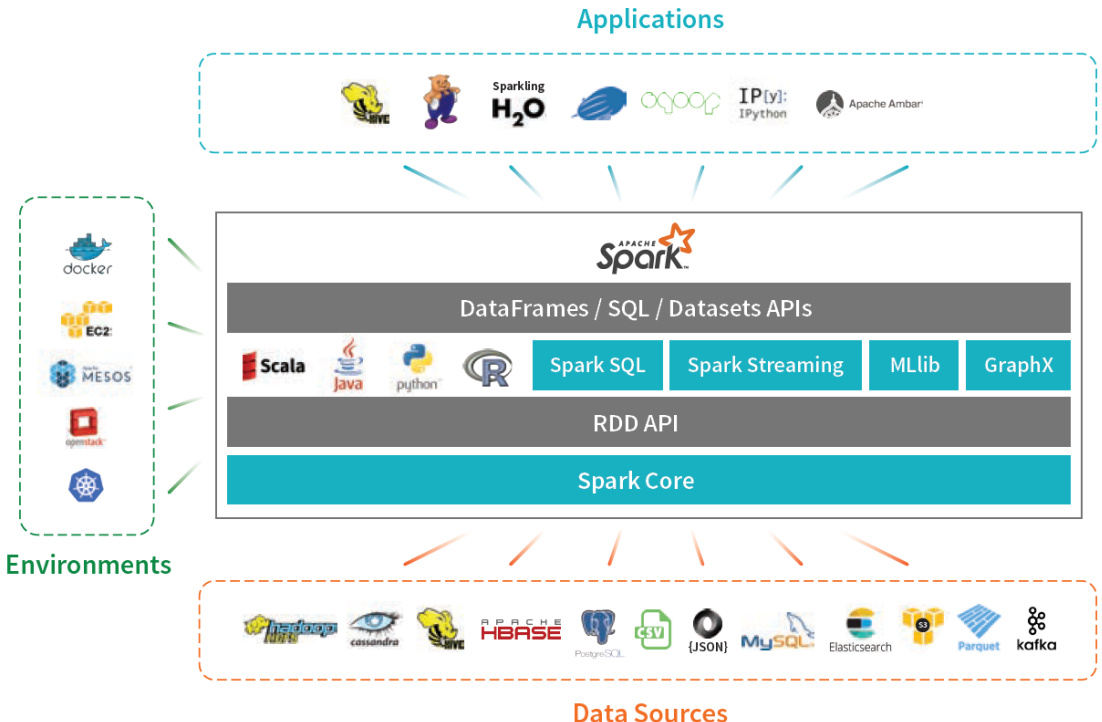
Extract-Transform-Load Systems, or Map-Reduce, or Big Data Frameworks

For: Large-scale, “ad hoc” data analysis

Mix of parallel and distributed architectures
Data usually coming from many different sources
Mix of SQL, Machine Learning, and ad hoc tasks (e.g., do image analysis, followed by SQL)

AWS Glue





Apache Spark

Okay…

- Key takeaway: Modern data architectures are all over the place
  - We haven't talked about NoSQL (MongoDB, etc.), Machine Learning, "Streaming"…
- Fundamentals haven't changed that much though
  - We are still either:
    - Going from some "input datasets" to an "output dataset" (queries/analytics)
    - Modifying data (transactions)
  - SQL is still very common, albeit often disguised
    - Spark RDD operations map nicely to SQL joins and aggregates (unified now)
    - MongoDB lookups, filters, and aggregates map to joins, selects, and aggregates in SQL
- But "performance trade-offs" are all over the place now
  - 30 years ago, we worried a lot about hard disks and things fitting in memory
  - Today, focus more on networks
- Focus has shifted to other aspects of data processing pipelines
  - Analytics/Machine learning, data cleaning, statistics

What we will cover…

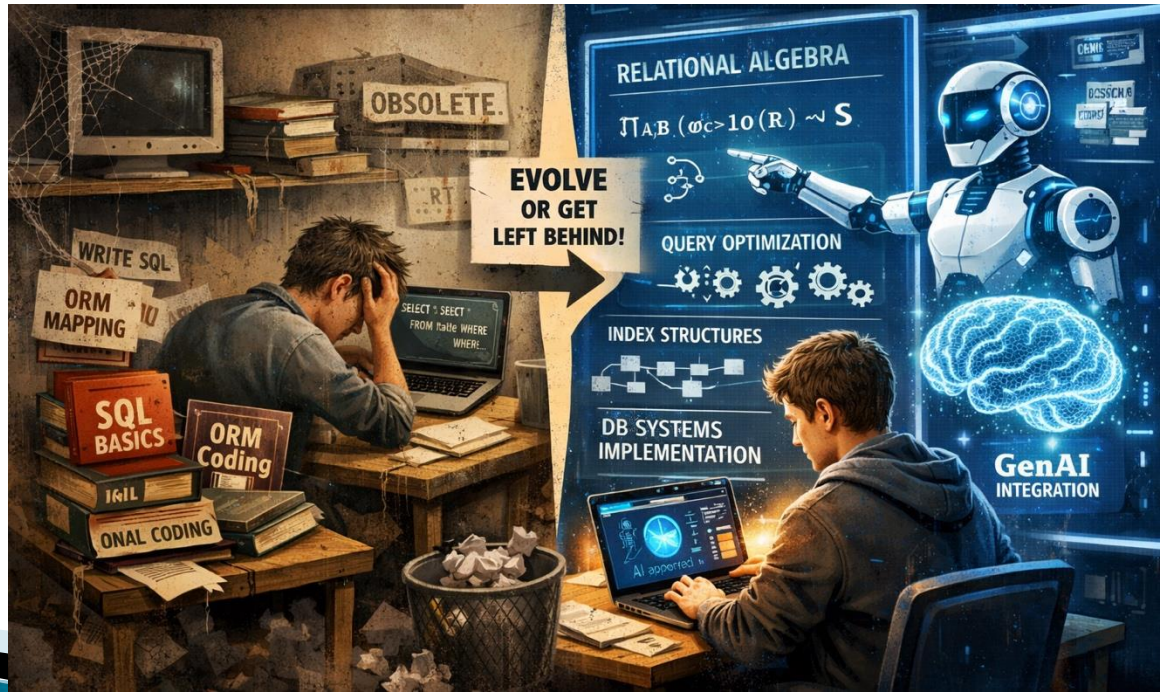
- Data Modeling
  - Data represented in tabular forms (Relational)
  - JSON/Document Data Model (MongoDB)
  - Entity-relationship Model for Schema Design
  - Normalization theory to formally define a ”good” schema
  - We will discuss a more primitive data model as well (Apache Spark)
- Query Languages and Programming Abstractions
  - SQL, MongoDB Query Language, Apache Spark MapReduce-like Framework, Some others
- Query Engines and Big Data Frameworks
  - Considerations in efficient implementations
- Transactions
  - How to reason about consistency and durability guarantees (ACID)
  - Basics of how to achieve those guarantees in centralized and distributed settings

What we will cover…

- This course is NOT about learning how to write SQL queries
  - Although we will practice that a fair bit…
- It is about learning the foundations of how data management systems are built and why they are built that way, so you:
  - Can think about the design decisions more systematically (e.g., through use of normalization theory and E-R modeling)
  - Understand the role of data modeling and how different models impact design decisions
  - Develop a deeper understanding of the differences and similarities between different types of database systems (e.g., PostgreSQL vs MongoDB)
  - Are equipped (to some extent) to design next-generation data systems
  - Are able to reason about performance of the queries/tasks

Structure of the Course

- Introduction
  - Motivation, data abstraction, common data systems architectures today
- Query Languages
  - Operations on datasets, and syntax for representing them
  - Cover: Relational Algebra, SQL, MongoDB Query Language (basics), and Spark (basics)
  - More advanced features of SQL
- Entity-relationship Models, ORMs, Normalization
  - How to create a database schema, and how to ensure it is “good”
- Implementation Issues
  - Storage and indexes
    - Different types of storage, and how to ensure reliability in presence of failures
    - Indexes for faster retrieval of data
  - Query processing and Optimization
    - How an SQL query is processed and optimized
  - Transactions
    - How to do concurrent updates correctly
    - How to ensure consistency in presence of failures

GenAI??

- No restrictions on use (except for proctored exams)
  - Final grades heavily exam-based
  - Exam questions will be very similar to homeworks
- Content: Plan to focus a bit more on formalisms, but mostly the same
  - Will provide guidance to use chatbots as AI Tutors for more in-depth learning
- Programming Assignments: Expecting at least some coding assistant use
  - Will have a path for students who don't want to use coding assistants